

Skin pathology

- Skin diseases are common and diverse
- Vary from inflammatory disorders to highly malignant neoplasms
- Many are intrinsic to the skin
- Some are manifestations of diseases involving many tissues
 - ❑ systemic lupus erythematosus
 - ❑ neurofibromatosis.
 - ❑ Acquired immunodeficiency syndrome (e.g., Kaposi sarcoma).

- The clinical history, gross appearance, and distribution of lesions are often as important as the microscopic findings in arriving at a specific diagnosis.

Macroscopic Lesions	Definition
Excoriation	Traumatic lesion breaking the epidermis and causing a raw linear area (i.e., deep scratch); often self-inflicted
Lichenification	Thickened, rough skin (similar to a lichen on a rock); usually the result of repeated rubbing
Macule, patch	Circumscribed, flat lesion distinguished from surrounding skin by color. Macules are 5 mm in diameter or less, while patches are greater than 5 mm in size.
Papule, nodule	Elevated dome-shaped or flat-topped lesion. Papules are 5 mm in diameter or less, while nodules are greater than 5 mm in size.
Plaque	Elevated flat-topped lesion, usually greater than 5 mm in diameter (may be formed by coalescence of papules)
Pustule	Discrete, pus-filled, raised lesion
Scale	Dry, horny, platelike excrescence; usually the result of imperfect cornification
Vesicle, bulla, blister	Fluid-filled raised lesion 5 mm or less in diameter (vesicle) or greater than 5 mm in diameter (bulla). <i>Blister</i> is the common term for both lesions.
Wheal	Itchy, transient, elevated lesion with variable blanching and erythema formed as the result of dermal edema

Microscopic Lesions	Definition
Acanthosis	Diffuse epidermal hyperplasia
Dyskeratosis	Abnormal, premature keratinization within cells below the stratum granulosum
Hyperkeratosis	Thickening of the stratum corneum, often associated with a qualitative abnormality of the keratin
Papillomatosis	Surface elevation caused by hyperplasia and enlargement of contiguous dermal papillae
Parakeratosis	Retention of nuclei in the stratum corneum of a squamous epithelium. On mucous membranes, parakeratosis is normal.
Spongiosis	Intercellular edema of the epidermis

- **Acantholysis:** Loss of intercellular connections resulting in loss of cohesion between keratinocytes
- **Papillomatosis:** Hyperplasia of the papillary dermis with elongation and/or widening of the dermal papillae
- **Lentiginous:** Refers to a linear pattern of melanocyte proliferation within the epidermal basal cell layer; lentiginous melanocytic hyperplasia can occur as a reactive change or as part of a neoplasm of melanocytes
- **Spongiosis:** Intercellular edema of the epidermis

Acute Inflammatory Dermatoses

- Acute lesions last from days to weeks and are characterized by inflammation marked by **mononuclear cells, edema, and (in some instances) epidermal, vascular, or subcutaneous injury.**
- Some acute lesions may persist, transitioning to a chronic phase, while others are self-limited

Urticaria

- Is a common disorder of the skin characterized by localized mast cell degranulation and resultant dermal microvascular hyperpermeability, culminating in pruritic edematous plaques called **wheals**.
- urticaria results from antigen-induced release of vasoactive mediators from mast cell granules via sensitization with specific immunoglobulin E (IgE) antibodies (type I hypersensitivity).
- IgE-dependent degranulation can follow exposure to a number of antigens (**pollens, foods, drugs, insect venom**).
- IgE-independent urticaria - substances that directly incite the degranulation of mast cells
 - ☐ opiates and
 - ☐ certain antibiotics.

- Lesions vary from small, pruritic papules to large edematous plaques.
- Sites of predilection for urticarial eruptions include any area exposed to pressure
 - ☐ The trunk
 - ☐ Distal extremities
 - ☐ Ears
- Sparse superficial perivenular infiltrate of mononuclear cells and rare neutrophils.
- Eosinophils may be present - midreticular dermis.
- Collagen bundles are more widely spaced.

- Urticaria typically affects individuals between **20 and 40 years of age**, but no age is immune.
- Individual lesions usually develop and fade within hours, but episodes can persist for days or even months
- Most cases respond to antihistamines, but more severe, refractory disease may require treatment with leukotriene antagonists, monoclonal antibodies that block the action of IgE, or immunosuppressive drugs.

Acute Eczematous Dermatitis

- **Eczema** is a clinical term that embraces a number of conditions with varied underlying etiologies.
- All are characterized by **red, papulovesicular, oozing, and crusted lesions** at an early stage.
- With persistence, lesions develop into **raised, scaling plaques**.

Classification of eczematous dermatitis:

- (1) Allergic contact dermatitis-delayed type hypersensitivity disease
 - (2) Atopic dermatitis
 - (3) Drug-related eczematous dermatitis
 - (4) Photoeczematous dermatitis
 - (5) Primary irritant dermatitis
- While atopic dermatitis reflects a genetic predisposition and can persist for years or decades, other forms of eczematous dermatitis resolve completely when the offending stimulus is removed or exposure is limited

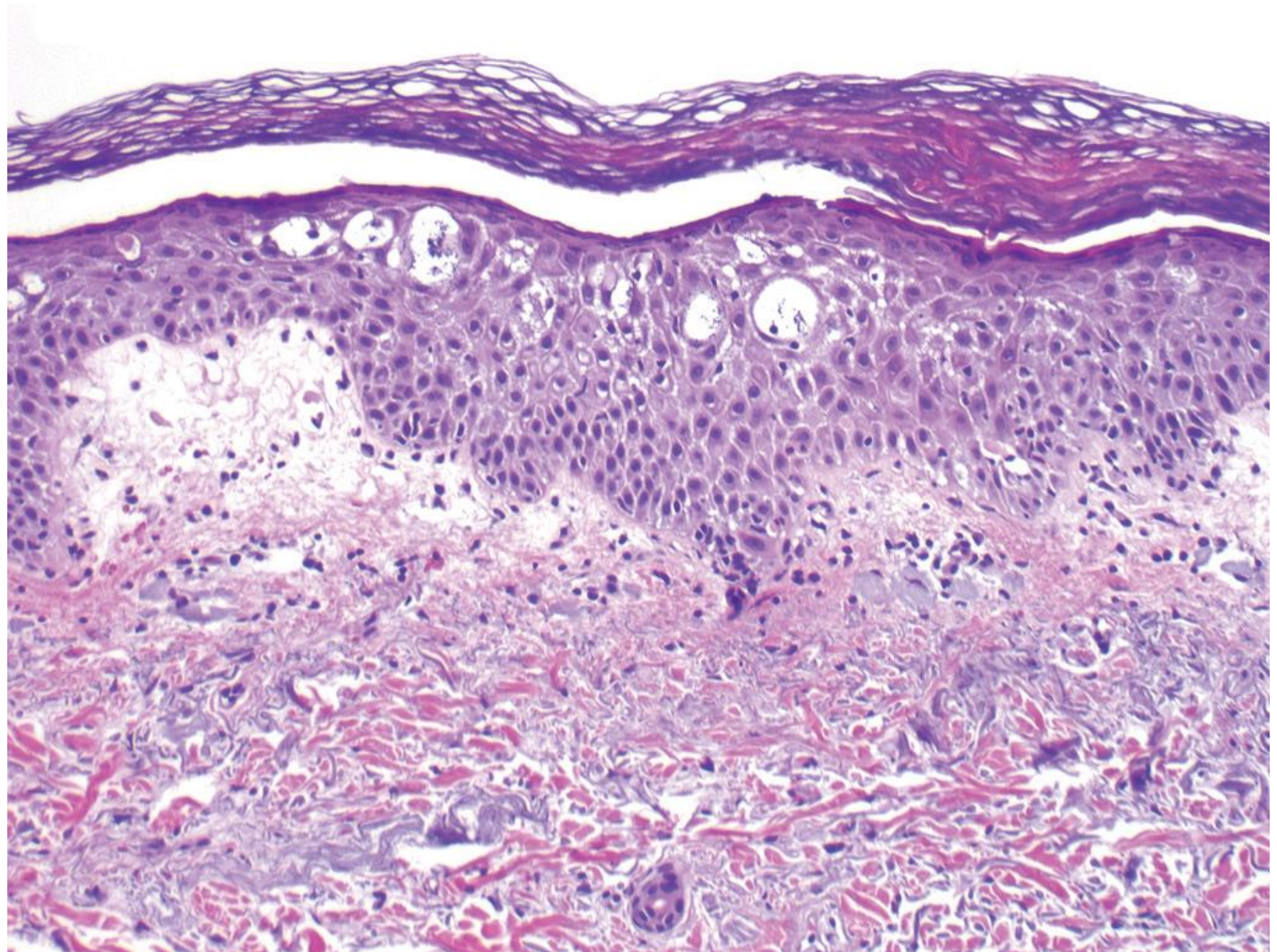
MORPHOLOGY

- **Spongiosis**-the accumulation of edema fluid within the epidermis.
- Intercellular bridges become stretched and more prominent >> "spongy" appearance to the epidermis.
- superficial, perivascular, lymphocytic infiltrate
- Papillary dermal edema and mast cell degranulation.
- Eosinophils may also be present

- **Allergic contact dermatitis** is triggered by exposure to an environmental contact-sensitizing agent, such as poison ivy, that chemically reacts with self-proteins, creating neoantigens that can be recognized by the T cell arm of the adaptive immune system.
- skin involvement is limited to sites of direct contact with the triggering agent ,whereas in other forms of eczema, lesions may be widely distributed.
- Susceptibility to atopic dermatitis is often inherited

- It usually appears in early childhood and remits spontaneously as patients mature into adults.
- Children with atopic dermatitis often have **asthma and allergic rhinitis**, termed the atopic triad.





Erythema Multiforme

- Erythema multiforme is characterized by epithelial injury mediated by skin-homing CD8+ cytotoxic T lymphocytes.
- It is an uncommon, usually self-limited disorder
- The cytotoxic T cell attack is focused on the basal cells of cutaneous and mucosal epithelia, presumably due to recognition of still unknown antigens.

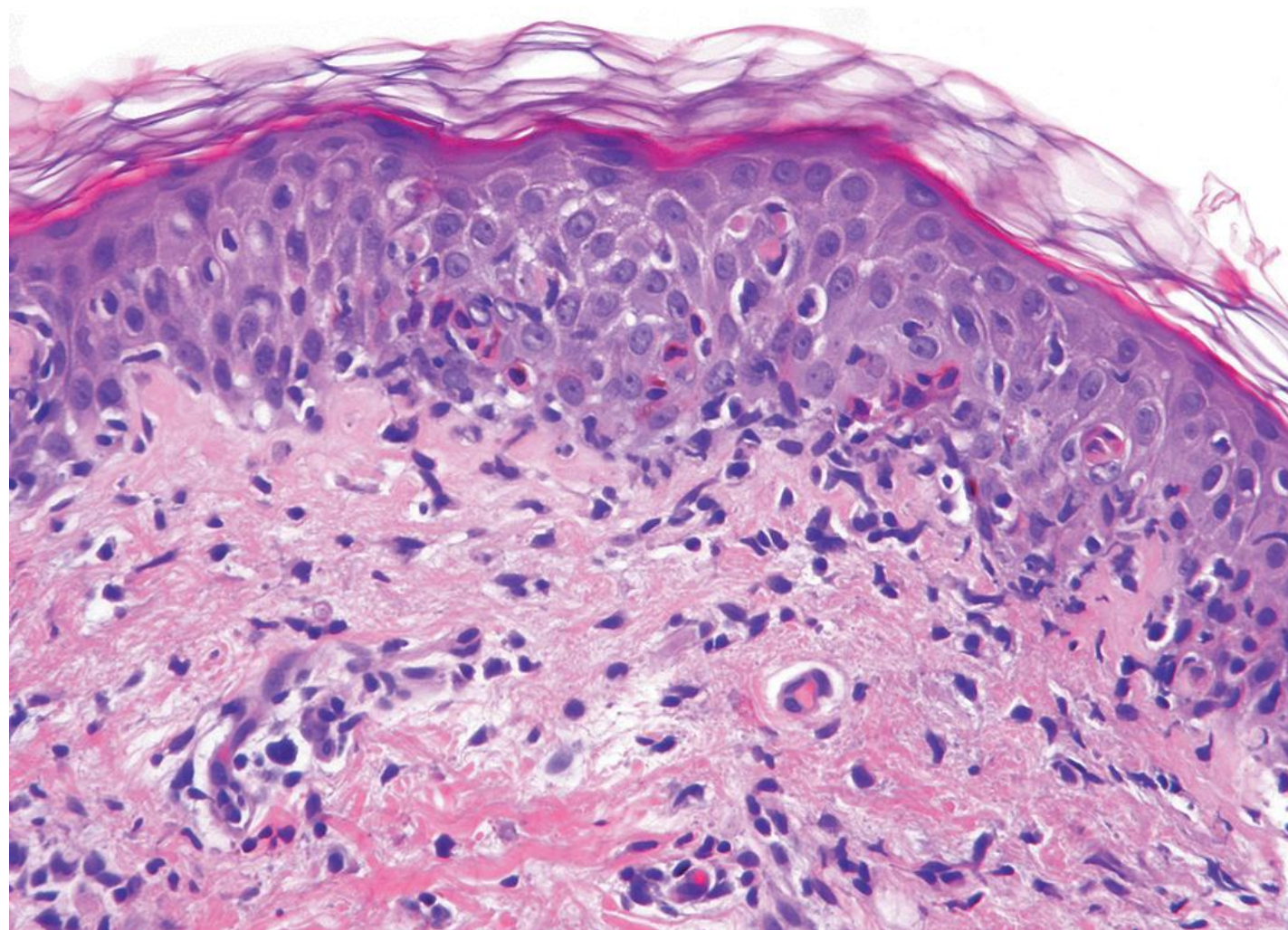
- It is associated with the following conditions:
- (1) infectious disorders (e.g., herpes simplex, Mycoplasma infections, histoplasmosis, coccidioidomycosis, typhoid, and leprosy);
- (2) administration of certain drugs (sulfonamides, penicillin, barbiturates, salicylates, hydantoins, and antimalarials);
- (3) malignancy (carcinomas and lymphomas); and
- (4) collagen vascular diseases (lupus erythematosus, dermatomyositis, and periarteritis nodosa).

- Lesions - **macules, papules, vesicles, and bullae**
- The characteristic **target lesion** consists of a red macule or papule with a pale vesicular or eroded center.
- An extensive and symptomatic febrile form of the disease, more common in children, is called the **Stevens-Johnson syndrome**.
- Erosions and crusting of the mucosal surfaces of the lips, conjunctiva, oral cavity, urethra, and anogenital region

- Superficial perivascular, lymphocytic infiltrate
- Dermal edema and
- Margination of lymphocytes along the dermoepidermal junction.
- Degenerating and necrotic keratinocytes.
- With time, discrete and confluent zones of epidermal necrosis occur, with concomitant blister formation.

- The forms associated with infection (most often herpesvirus) are less severe.
- Erythema multiforme caused by medications may progress to **more serious eruptions**, such as **Stevens-Johnson syndrome or toxic epidermal necrolysis**.
- These forms can be life-threatening, as they may cause sloughing of large portions of the epidermis, resulting in **fluid loss and infections complications** similar to those seen in burn-injured patients.





CHRONIC INFLAMMATORY DERMATOSES

- Chronic inflammatory dermatoses are persistent skin conditions that exhibit their most characteristic features over many months to years
- Roughened skin as a result of excessive or abnormal scale formation and shedding (desquamation).

Psoriasis

- Psoriasis is a common chronic inflammatory dermatosis
- Sometimes associated with arthritis, myopathy, enteropathy, spondylitic heart disease, and the acquired immunodeficiency syndrome.
- The most typical lesion is a **well-demarcated, pink to salmon-colored plaque** covered by loosely adherent scales that are characteristically **silver-white**.
- Nail changes occur in 30% of cases of psoriasis and consist of **yellow-brown discoloration**.

- Psoriasis is a T cell-mediated inflammatory disease, presumed to be autoimmune in origin, although the antigens are not well described.
- Seems to be multifactorial - genetic susceptibility and environmental agents.
- Sensitized populations of T cells home to the dermis, including **CD4+ TH17 and TH1 cells and CD8+ T cells**, and accumulate in the epidermis.
- T cells infiltrating the skin may create an abnormal microenvironment by secreting cytokines and growth factors that influence keratinocyte replication and senescence pathways, resulting in the characteristic **inflammatory and proliferative lesions**.

- Acanthosis with regular downward elongation of the rete ridges.
- The stratum granulosum is thinned or absent,
- Extensive overlying parakeratotic scale
- Thinning of the portion of the epidermal cell layer that overlies the tips of dermal papillae and dilated, tortuous blood vessels within these papillae.
- These vessels bleed readily when the scale is removed, giving rise to multiple punctate bleeding points (**Auspitz sign**).
- Neutrophils form small aggregates within parakeratotic stratum corneum -**Munro microabscesses**.

- Psoriasis most frequently affects the skin of the elbows, knees, scalp, lumbosacral areas, intergluteal cleft, glans penis, and vulva.
- Nail changes on the fingers and toes occur in 30% of cases.





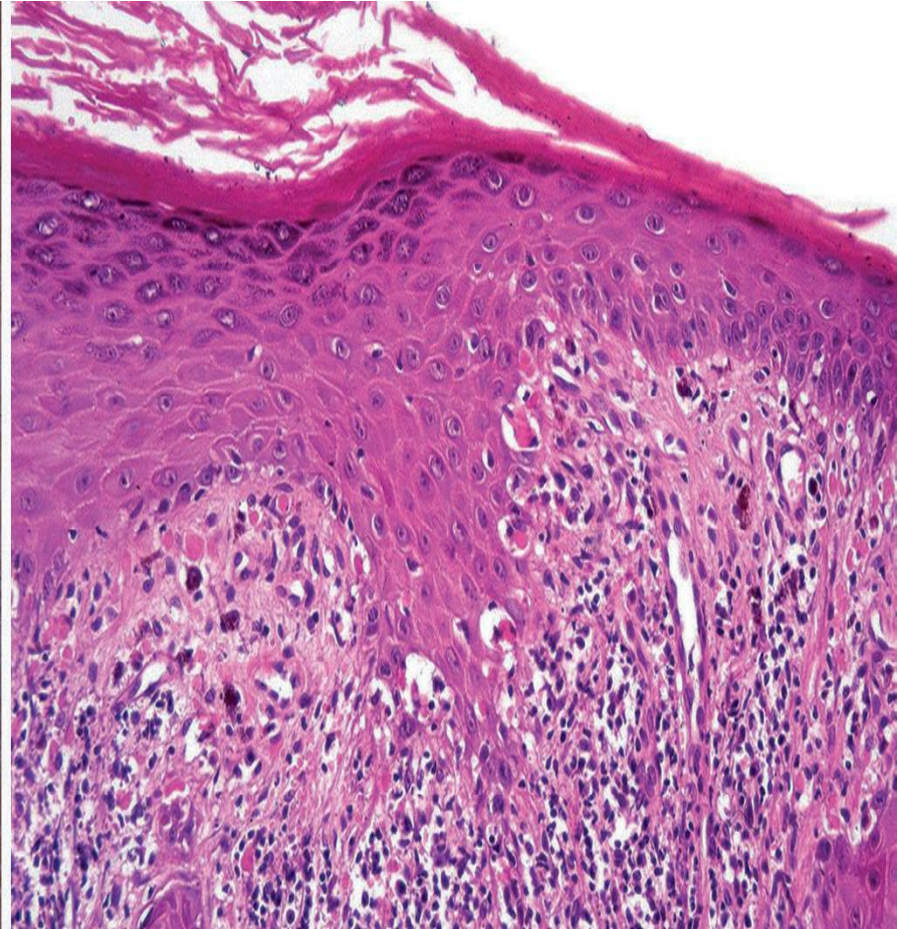
- The clinical subtypes are defined by pattern of involvement and severity.
- Treatment is aimed at preventing the release or actions of inflammatory mediators.
- Mild disease is treated topically with ointments containing **corticosteroids** or other immunomodulatory agents
- More severe disease is treated with phototherapy (which has immunosuppressive effects) or systemic therapy with immunosuppressive agents such as methotrexate or TNF antagonists.

Lichen Planus

- "Pruritic, purple, polygonal papules"
- The lesions may result from a CD8+ T cell–mediated cytotoxic response against antigens in the basal cell layer and the dermoepidermal junction that are produced by unknown mechanisms, perhaps as a consequence of a **viral infection or drug exposure**
- Disorder of skin and mucous membranes.
- Self-limited and resolves spontaneously 1 to 2 years after onset, leaving post inflammatory hyperpigmentation.
- Cutaneous lesions - itchy, violaceous, flat-topped papules, may coalesce focally to form plaques.

MORPHOLOGY

- These papules are highlighted by white dots or lines termed Wickham striae.
- Hyperpigmentation may result from melanin loss into the dermis from damaged keratinocytes.
- Continuous infiltrate of lymphocytes along the dermoepidermal junction.
- Basal keratinocytes - degeneration, necrosis.
- Pattern of inflammation causes the dermoepidermal interface to assume an angulated, zigzag contour - "sawtoothing".
- Epidermal hyperplasia
- Thickening of the granular cell layer and stratum corneum (hypergranulosis and hyperkeratosis).



- The precise pathogenesis of lichen planus is not known.
- Release of antigens at the levels of the basal cell layer and the dermoepidermal junction may elicit a cell-mediated cytotoxic immune response.

INFECTIOUS DERMATOSES

- Bacterial Infections

Impetigo

- One of the most common bacterial infections of the skin, is seen primarily in children.
- The causative organism is usually **Staphylococcus aureus or, less commonly, Streptococcus pyogenes**
- Typically acquired through direct contact with a source.
- Impetigo often begins as a single small macule, usually on the extremities or the face near the nose or the mouth, which rapidly evolves into a larger lesion
- characterized by an accumulation of neutrophils beneath the stratum corneum that often produces a subcorneal pustule.

Fungal Infections

- Fungal infections can be:
 - ❑ Superficial (stratum corneum, hair, and nails)
 - ❑ Deep (dermis or subcutis)
 - ❑ Systemic (arising through hematogenous spread, often in an immunocompromised patient)

Morphology

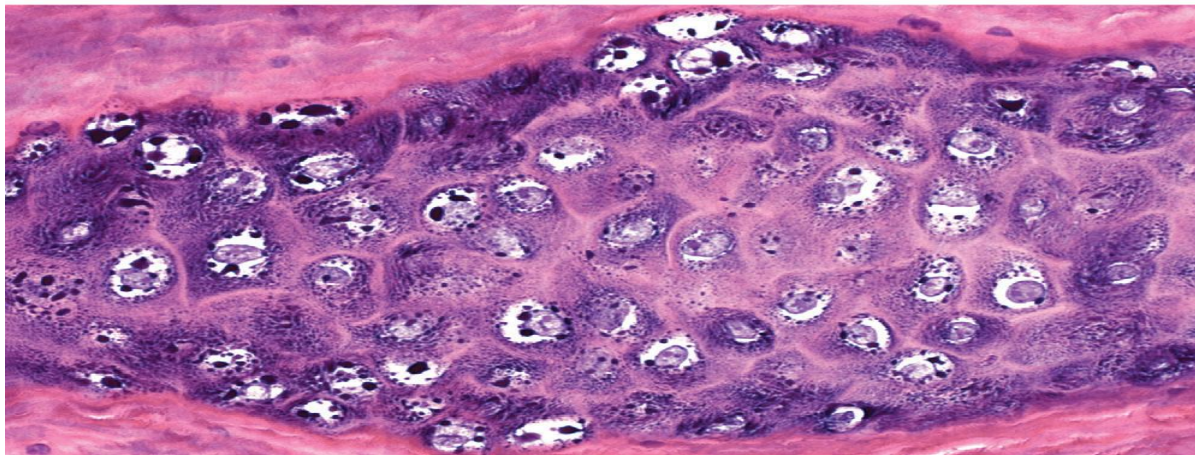
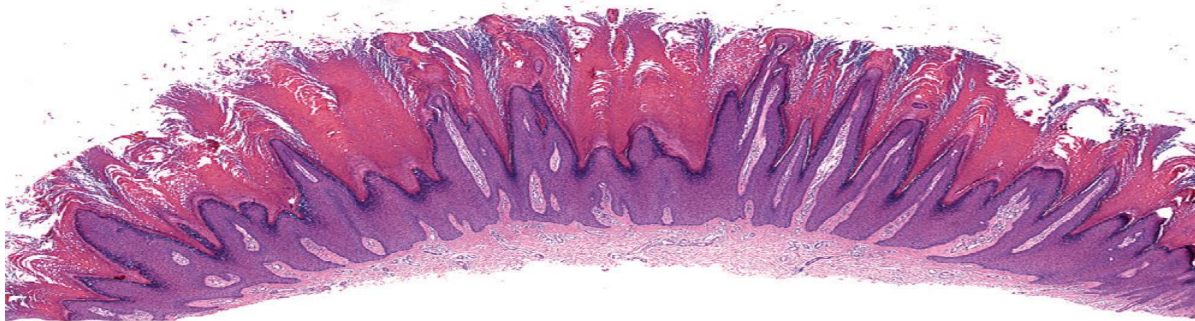
- Superficial infections are often associated with a neutrophilic infiltrate in the epidermis.
- Deep fungal infections produce greater tissue damage and often elicit a granulomatous response.
- Aspergillus can be angioinvasive.
- Periodic acid–Schiff (PAS) and Gomori methenamine silver stains are helpful in identifying the fungal organisms.

Verrucae (Warts)

- Verrucae are proliferative lesions of squamous epithelial cells that are caused by human papillomavirus (HPV).
- HPV infection usually stems from direct contact with an infected individual or autoinoculation.
- Verrucae generally are self-limited, most often regressing spontaneously within 6 months to 2 years.

- Why low-risk viruses cause warts instead of cancer is likely due to structural variation in E6 and E7 proteins that affect their interactions with host proteins.
- Because the growth of warts is normally halted by the immune response, immunodeficiency is associated with more numerous and larger verrucae.

- Histologic features common to verrucae include epidermal hyperplasia, which is often undulant in character (so-called verrucous or papillomatous epidermal hyperplasia)
- Cytoplasmic vacuolization (koilocytosis), which preferentially involves the more superficial epidermal layers, producing halos of pallor surrounding infected nuclei.



Blistering (Bullous) Disorders

- group of disorders in which blisters are the primary and most distinctive feature.
- Blistering in these diseases tends to occur at specific levels within the skin, a morphologic distinction that is critical for diagnosis

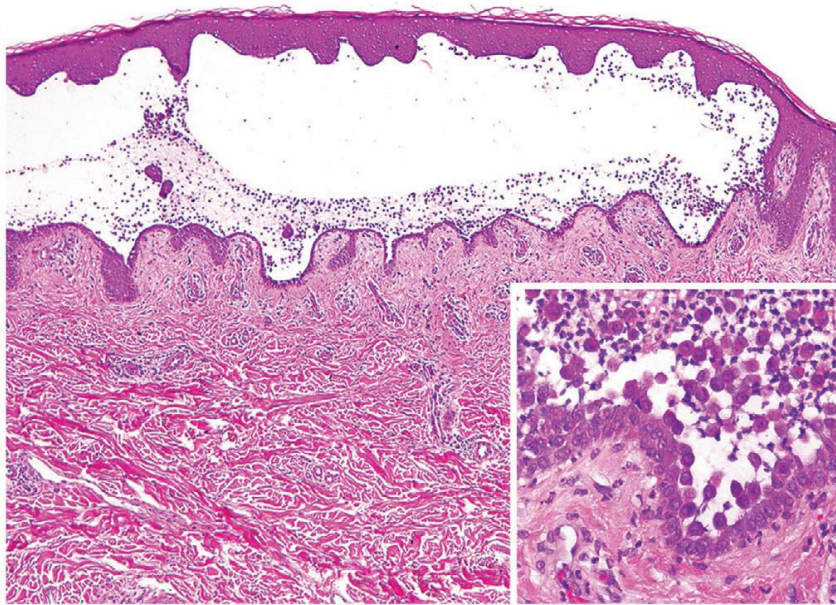
- **Pemphigus (Vulgaris and Foliaceus)**
- Pemphigus is an uncommon autoimmune blistering disorder resulting from loss of normal intercellular attachments within the epidermis and the squamous mucosal epithelium.
- There are three major variants:
 - ❑ Pemphigus vulgaris (the most common type)
 - ❑ Pemphigus foliaceus
 - ❑ Paraneoplastic pemphigus

- Pemphigus vulgaris and pemphigus foliaceus are caused by antibody-mediated (type II) hypersensitivity reactions
- The pathogenic antibodies are IgG autoantibodies that bind to intercellular desmosomal proteins (desmoglein types 1 and 3)

- Pemphigus vulgaris involves both mucosa and skin, especially on the scalp, face, axillae, groin, trunk, and points of pressure.
- The lesions are superficial flaccid vesicles and bullae that rupture easily, leaving deep and often extensive erosions covered with a serum crust

- Pemphigus foliaceus, a rare, milder form of pemphigus, results in bullae that are mainly confined to the skin, with only infrequent involvement of mucous membranes.
- The blisters in this disorder are superficial, such that more limited zones of erythema and crusting of ruptured blisters are seen

- In pemphigus vulgaris, acantholysis selectively involves the layer of cells immediately above the basal cell layer, giving rise to a **suprabasal acantholytic blister**
- In pemphigus foliaceus, acantholysis selectively involves the superficial epidermis at the level of the **stratum granulosum**





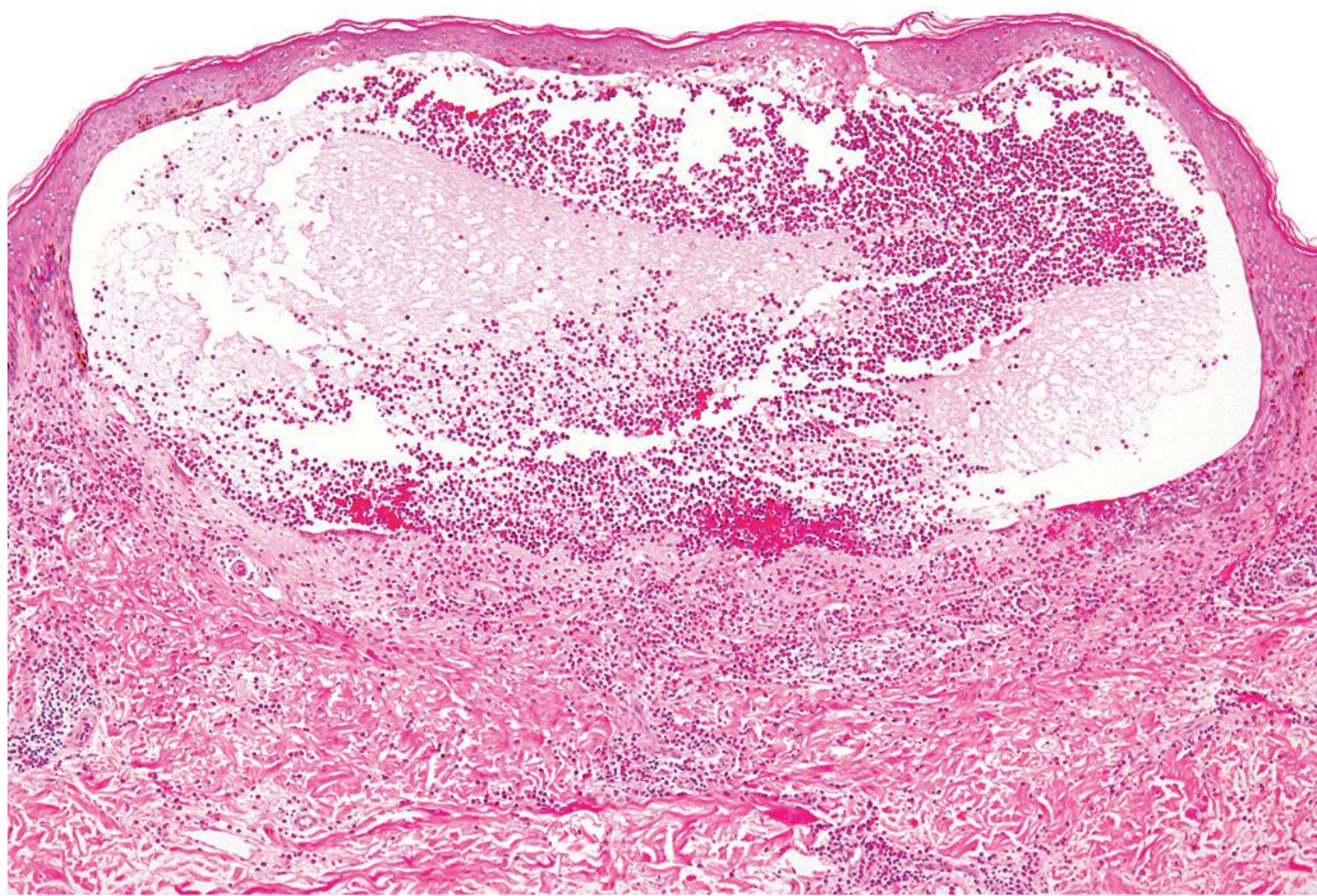
Bullous Pemphigoid

- Blistering in bullous pemphigoid is triggered by the linear deposition of autoreactive IgG antibodies and complement in the epidermal basement membrane
- The proteins that are recognized by the autoantibodies have structural roles in dermoepidermal adhesion.

- Bullous pemphigoid is associated with tense subepidermal bullae filled with clear fluid
- The overlying epidermis characteristically lacks acantholysis.
- Early lesions show variable numbers of eosinophils at the dermal-epidermal junction, occasional neutrophils, superficial dermal edema, and associated basal cell layer vacuolization.
- The vacuolated basal cell layer eventually gives rise to a fluid-filled blister
- The blister roof consists of full-thickness epidermis with intact intercellular junctions

- The lesions of bullous pemphigoid do not rupture as readily as in pemphigus and, if uncomplicated by infection, heal without scarring.
- The disease tends to follow a remitting and relapsing course and responds to topical or systemic immunosuppressive agents





Dermatitis Herpetiformis

- autoimmune blistering disorder associated with gluten sensitivity that is characterized by **extremely pruritic grouped vesicles and papules**.
- Up to 80% of cases are associated with celiac disease
- IgA antibodies to dietary gluten (derived from the wheat protein gliadin) as well as IgA autoantibodies that cross-react with endomysium and tissue transglutaminases, including epidermal transglutaminase expressed by keratinocytes

- The lesions of dermatitis herpetiformis are bilateral, symmetric, and grouped and preferentially involve the **extensor surfaces, elbows, knees, upper back, and buttocks**



Freckles

- Freckles are the most common pigmented lesions of childhood in lightly pigmented individuals.
- Freckles are generally small (1 to several mm in diameter), tan-red or light brown macules that appear after sun exposure.
- Once present, freckles fade and darken in a cyclic fashion during winter and summer, respectively.
- This is not because of changes in the number of melanocytes, but in the degree of pigmentation.
- Hyperpigmentation of freckles results from increased amounts of melanin pigment within basal keratinocytes.
- Associated melanocytes may be slightly enlarged but are normal in density

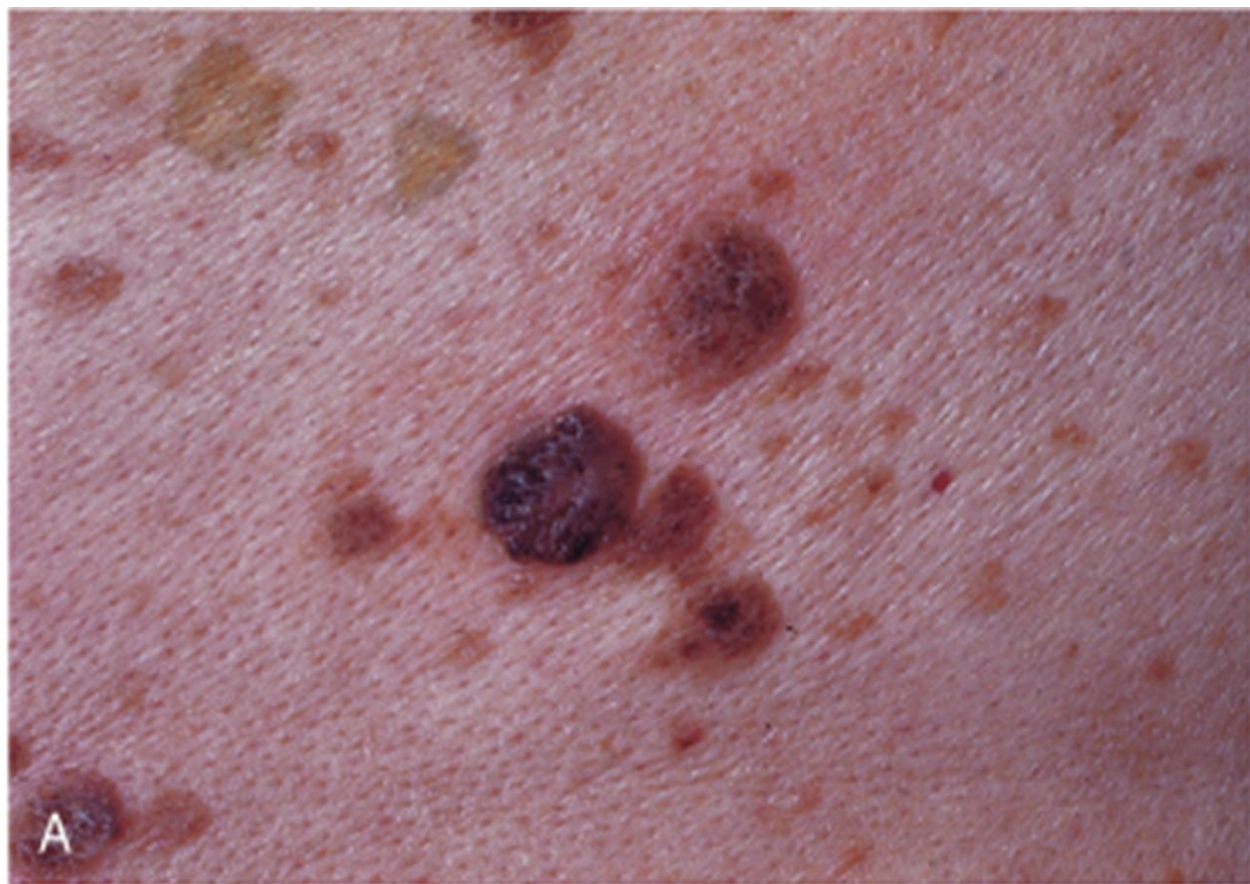
lentigo

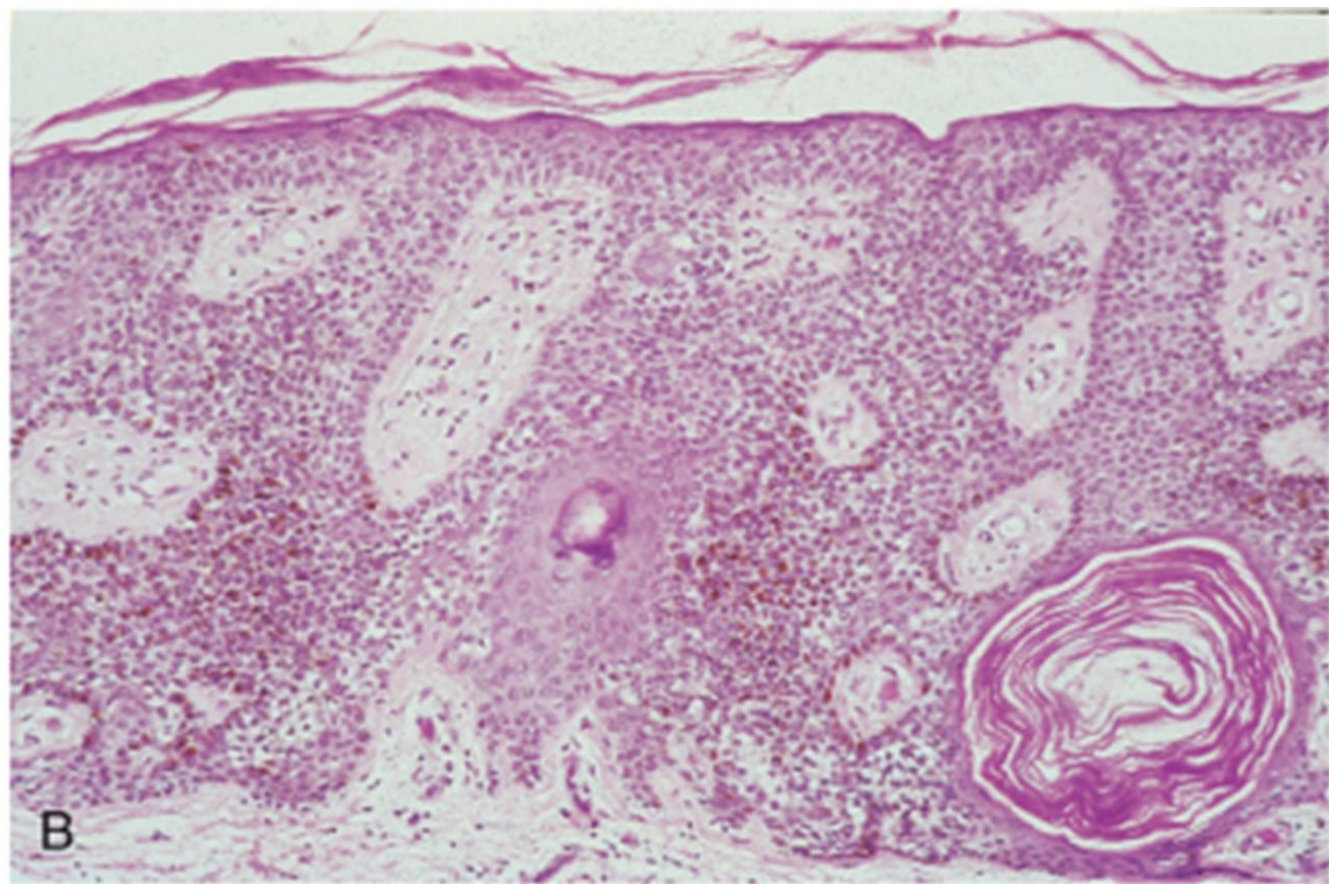
- The term lentigo (plural, lentigines) refers to a common benign localized hyperplasia of melanocytes occurring at all ages, but often initiated in infancy and childhood.
- There is no sex or racial predilection, and the cause and pathogenesis are unknown.
- Lentigines may involve mucous membranes as well as the skin and consist of small (5 to 10 mm across), oval, tan-brown macules or patches.
- The essential histologic feature is linear (nonnested) melanocytic hyperplasia restricted to the cell layer immediately above the basement membrane that produces a hyperpigmented basal cell layer.

SEBORRHEIC KERATOSIS

- Clinically appear as round, flat, coinlike plaques that vary in diameter from millimeters to several centimeters.
 - are uniformly tan to dark brown and usually show a velvety to granular surface.
 - Occasionally, become inflamed or mimic melanoma.
 - UV light
 - Genetics
- risk

- Histologically - exophytic and are demarcated sharply from the adjacent epidermis.
- composed of sheets of small cells that resemble basal cells of the normal epidermis.
- Melanin pigmentation is present within these basaloid cells - brown coloration.
- Exuberant keratin production (hyperkeratosis) occurs at the surface of seborrheic keratoses,
- Small keratin-filled cysts (horn cysts).





KERATOACANTHOMA

- is a rapidly developing neoplasm that clinically and histologically may mimic well-differentiated squamous cell carcinoma, but it heals spontaneously without treatment.
- Appear as flesh-colored, dome-shaped nodules with a central, keratin-filled plug(crater).

- Histologically - central keratin-filled crater surrounded by proliferating epithelial cells that extend upward in a liplike fashion over the sides of the crater and downward into the dermis as irregular tongues.

ACTINIC KERATOSIS

- a series of progressively dysplastic changes occur before the development of overt malignancy.
- Because this dysplasia in the skin is usually the result of chronic exposure to sunlight and is associated with build-up of excess keratin
 - Actinic keratoses
- Lesions of actinic keratosis are usually less than 1 cm in diameter; are tan-brown, red, or skin colored; and have a rough, sandpaper-like consistency

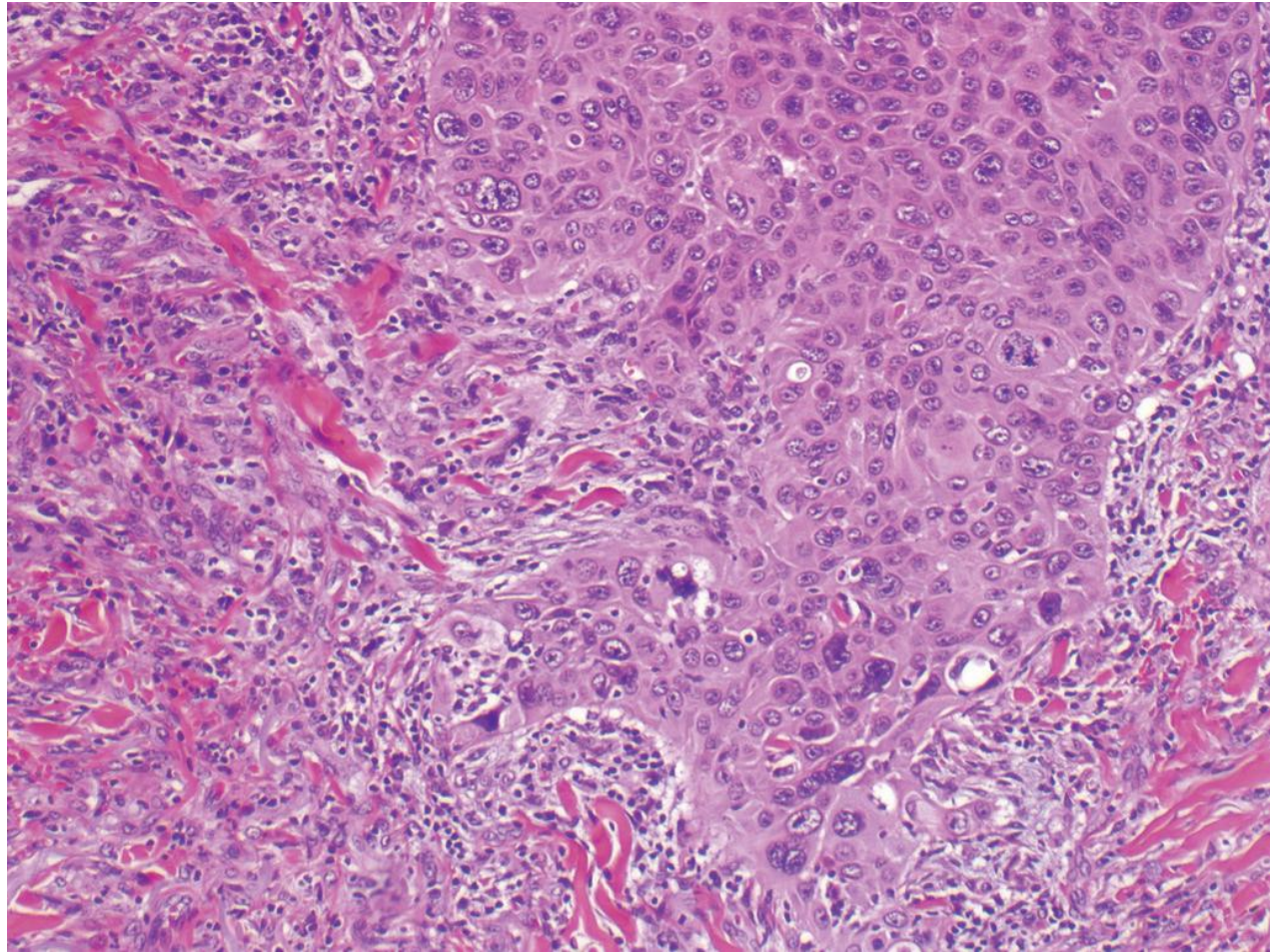
- Cytologic atypia in the lowermost layers of the epidermis.
- The dermis contains thickened, blue-gray elastic fibers (elastosis) - abnormal dermal elastic fiber synthesis by sun-damaged fibroblasts within the superficial dermis.
- The stratum corneum is thickened
- Parakeratosis
- Some lesions progress to full-thickness atypia, squamous cell carcinoma in situ.

SQUAMOUS CELL CARCINOMA

- The most common tumor arising on sun-exposed sites in older people.
- Predisposing factors - **sunlight, industrial carcinogens (tars and oils), chronic ulcers, draining osteomyelitis, old burn scars, ingestion of arsenicals, ionizing radiation, and tobacco and betel nut chewing.**
- In situ carcinoma - sharply defined, red, scaling plaques.
- Advanced invasive lesions - nodular, variable keratin production(hyperkeratosis) and may ulcerate

- Squamous cell carcinoma in situ is characterized by highly atypical cells at all levels of the epidermis
- Invasive squamous cell carcinomas exhibit variable differentiation

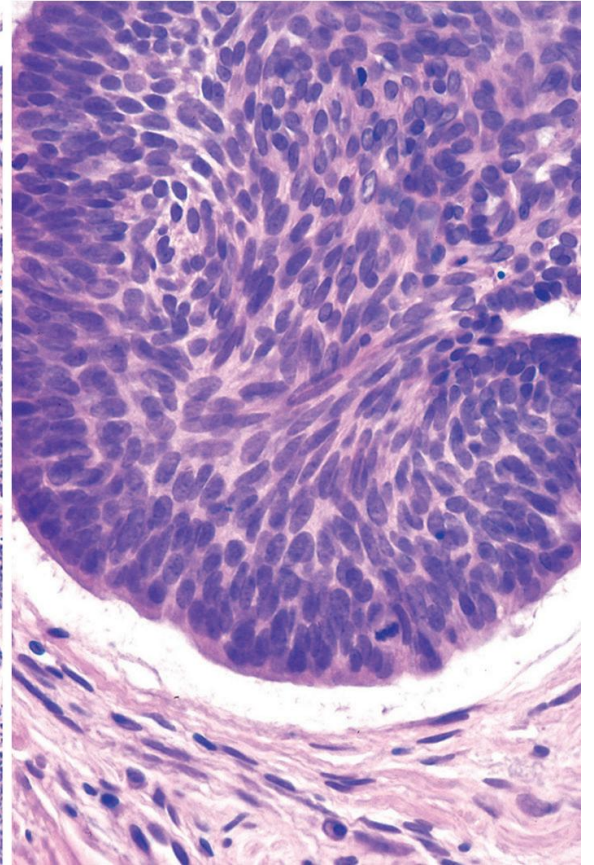
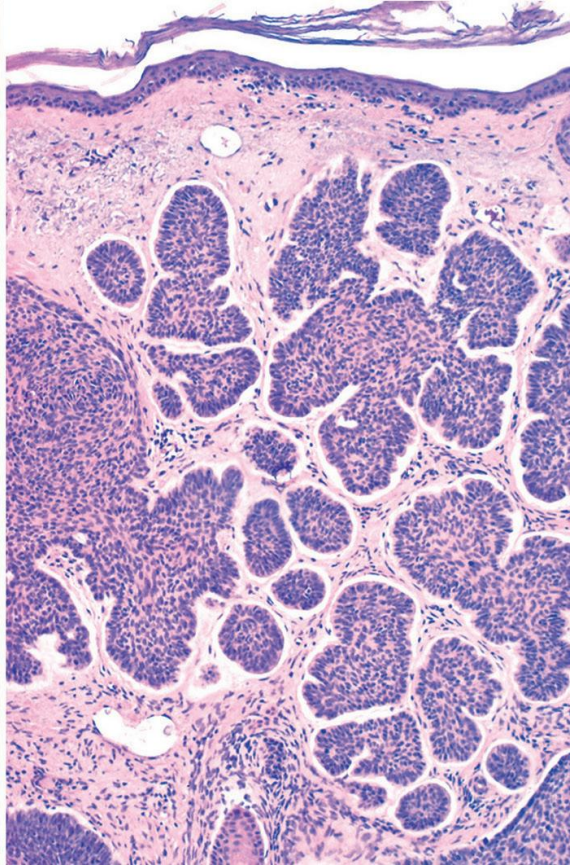




BASAL CELL CARCINOMA

- Slow-growing tumors that rarely metastasize.
- They have a tendency to occur at sites subject to chronic sun exposure and in lightly pigmented people.
- Clinically, pearly papules, often containing prominent, dilated subepidermal blood vessels (telangiectasia).
- Some tumors contain melanin pigment and appear similar to nevocellular nevi or melanomas.
- Advanced lesions may ulcerate, and extensive local invasion of bone or facial sinuses -"rodent ulcers."

- Histologically, tumor cells resemble those in the normal basal cell layer of the epidermis.
- Cords and islands of variably basophilic cells with hyperchromatic nuclei, embedded in a mucinous matrix, and often surrounded by many fibroblasts and lymphocytes.
- The cells forming the periphery of the tumor cell islands tend to be arranged radially with their long axes in approximately parallel alignment (**palisading**).



NEVOCELLULAR NEVUS (PIGMENTED NEVUS, MOLE)

- Nevocellular nevus refers to any congenital or acquired neoplasm of melanocytes.
- Clinically, common acquired nevocellular nevi are tan-to-brown, uniformly pigmented, small (usually 5 mm or less across), solid regions of elevated skin (papules) with well-defined, rounded borders.

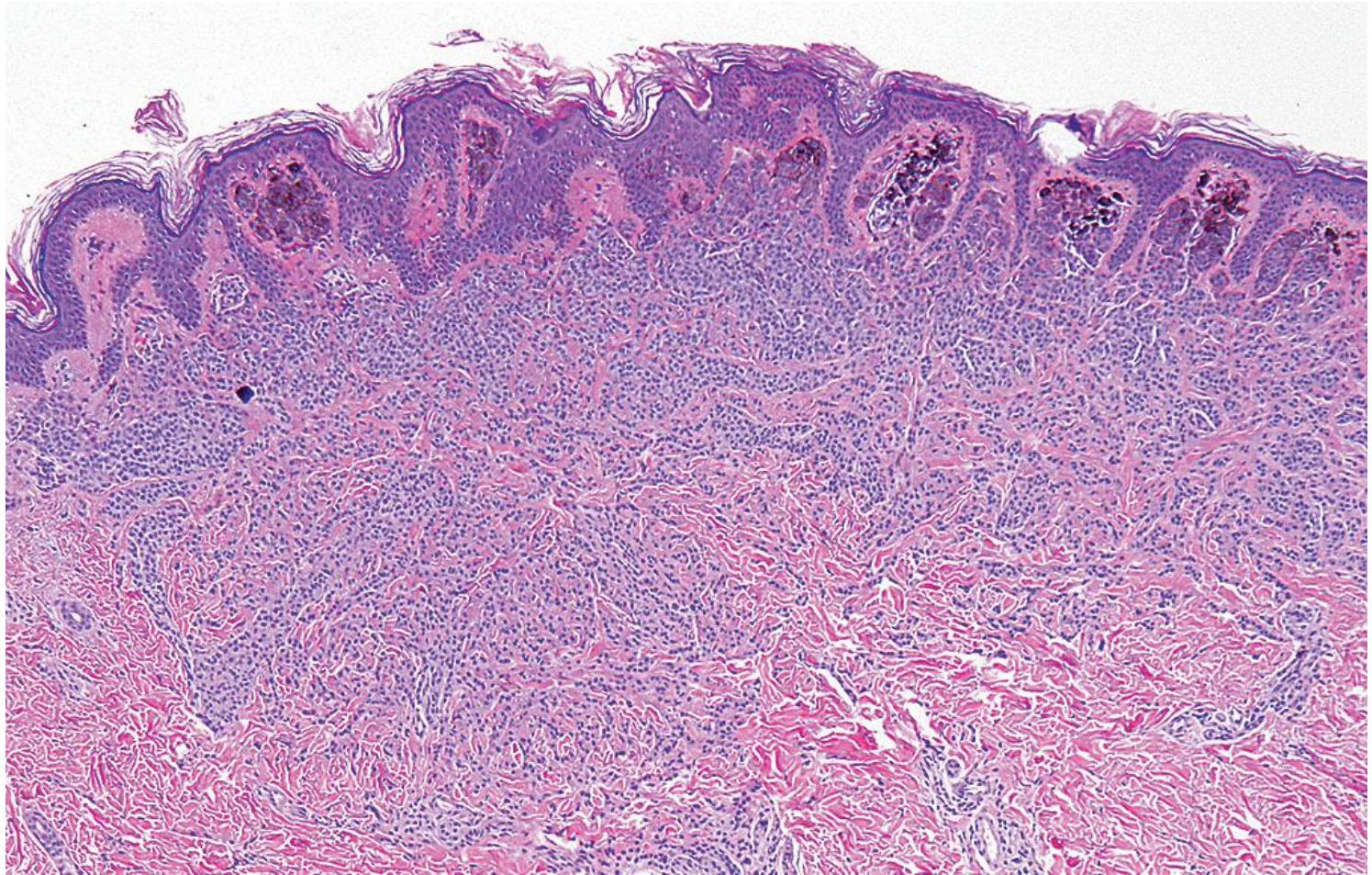
- Formed by melanocytes that have been transformed from highly dendritic single cells, normally interspersed among basal keratinocytes, to **round-to-oval cells that grow in aggregates, or "nests," along the dermoepidermal junction.**
- Eventually, most junctional nevi grow into the underlying dermis as nests or cords of cells (**compound nevi**), and in older lesions the epidermal nests may be lost entirely, creating **intradermal nevi**

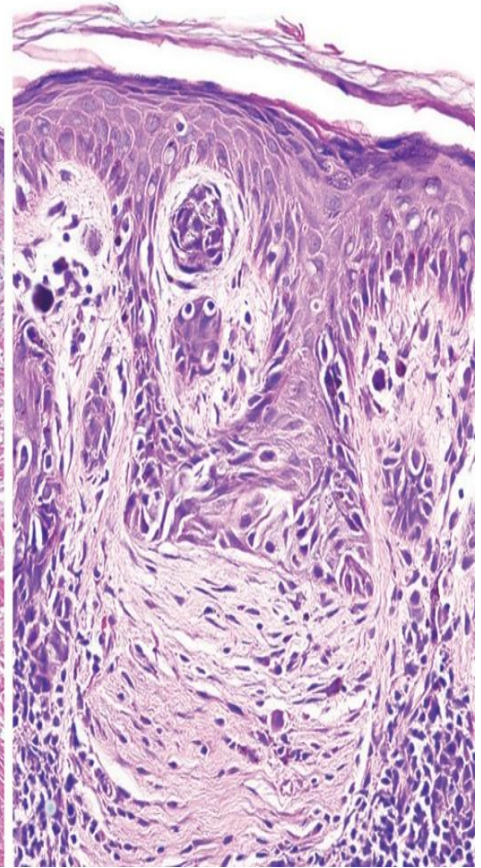
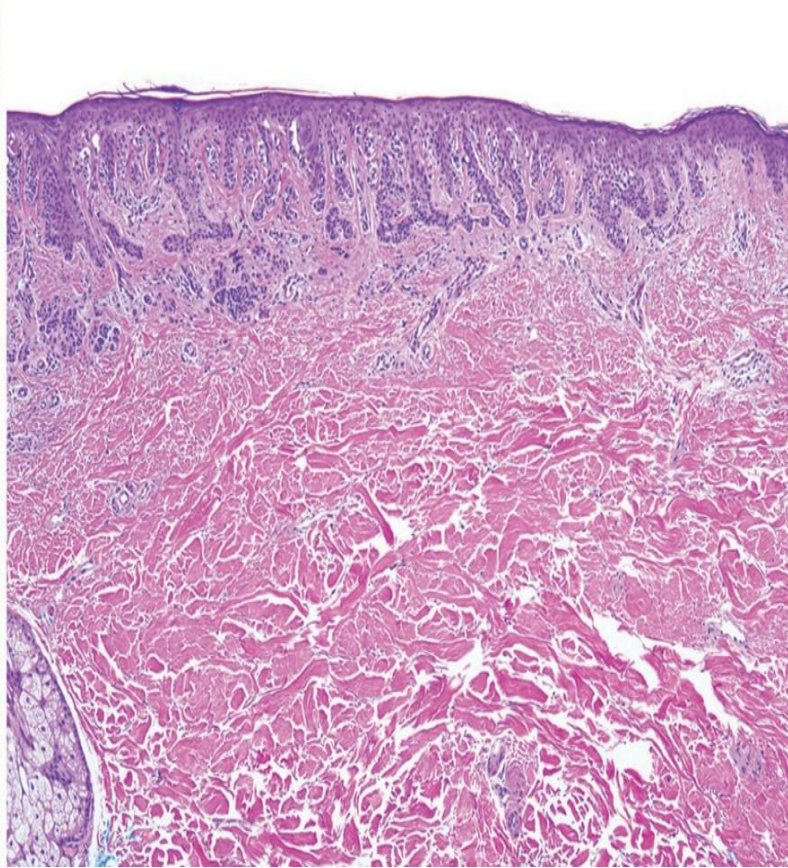
DYSPLASTIC NEVI

- Dysplastic nevi may occur sporadically or in a familial form.
- Clinically, dysplastic nevi are larger than most acquired nevi (often more than 5 mm across) and may occur as hundreds of lesions on the body surface.
- They are flat macules to slightly raised plaques
- Unlike ordinary moles, dysplastic nevi have a tendency to occur on non-sun-exposed as well as sun-exposed body surfaces.

- Histologically, dysplastic nevi consist of compound nevi with both architectural and cytologic evidence of abnormal growth.
- Nevus cell nests within the epidermis may be enlarged and exhibit abnormal fusion or coalescence with adjacent nests.
- Single nevus cells begin to replace the normal basal cell layer along the dermoepidermal junction, producing so-called **lentiginous hyperplasia**.
- Cytologic atypia consisting of irregular, often angulated, nuclear contours and hyperchromasia is frequently observed.







MALIGNANT MELANOMA

- Most of these lesions arise in the skin
- Other sites of origin –
 - ❑ Oral and anogenital mucosal surfaces,
 - ❑ Esophagus,
 - ❑ Meninges, and
 - ❑ Eye

- Sunlight plays an important role in the development of malignant melanoma.
- The incidence is highest in sun-exposed skin and in geographic locales.
- The presence of a preexisting nevus (e.g., a dysplastic nevus), hereditary factors, or even exposure to certain carcinogens may play a role in lesion development and evolution.

- Clinically, malignant melanoma of the skin is usually **asymptomatic, although itching may be an early manifestation.**
- The most important clinical sign of the disease is **a change in the color or size in a pigmented lesion.**
- Melanomas exhibit striking **variations in pigmentation, appearing in shades of black, brown, red, dark blue, and gray.**
- The borders of melanomas are **irregular and often "notched."**

- The most important warning signs, sometimes called the ABCDEs of melanoma, are
- (1) asymmetry
- (2) irregular borders
- (3) variegated color
- (4) increasing diameter
- (5) evolution or change over time, especially if rapid.

- The clinical warning signs of melanoma are
- (1) enlargement of a preexisting mole,
- (2) itching or pain in a preexisting mole,
- (3) development of a new pigmented lesion during adult life,
- (4) irregularity of the borders of a pigmented lesion, and
- (5) variegation of color within a pigmented lesion.

- Metastases involve not only regional lymph nodes but also liver, lungs, brain, and virtually any other site that can be seeded by the hematogenous route.

- Individual melanoma cells are usually considerably larger than nevus cells.
- Large nuclei with irregular contours having chromatin clumped at the periphery of the nuclear membrane and prominent red (eosinophilic) nucleoli.
- Cells grow as poorly formed nests or as individual cells at all levels of the epidermis and, in the dermis, as expansile, balloon-like nodules.

- Increasing thickness strongly correlates with worse biologic behavior of melanomas (termed **Breslow thickness**).
- The vast majority of superficial lesions are curable surgically, while metastatic melanoma has a very poor prognosis

